

# SCD\_Test\_Modify\_Staging\_CustomerNotebook

New notebook

Generate the sample staging customer data to test the SCD

**Generating this synthetic customer data as template for our production template for SCD Type handling of our dimensions**

Because we have two separate areas, here we generate the data to the staging, and in Slowly Changing Dim Notebooks, we can run the **MERGE**

We can change values to test here, and then go to the other notebook to test

```
In [1]: # Welcome to your new notebook
# Type here in the cell editor to add code!
from pyspark.sql import SparkSession
import pandas as pd
import numpy as np
from datetime import datetime, timedelta
```

StatementMeta(, 9d98d0b2-e722-4853-81cd-23f0968cf40f, 3, Finished, Available, Finished, False)

## Modify Tracked Columns for SCD Type 2 Test

**SCD Type 2 Data manipulation test - varying values**

```
In [2]: import random

num_changes = 20
rows_to_change = random.sample(range(200), num_changes)
```

StatementMeta(, 9d98d0b2-e722-4853-81cd-23f0968cf40f, 4, Finished, Available, Finished, False)

Modify the columns tracked by SCD

**E.g. SCD columns are:**

```
In [3]: scd_columns = ["customer_name", "customer_status", "address"]
```

StatementMeta(, 9d98d0b2-e722-4853-81cd-23f0968cf40f, 5, Finished, Available, Finished, False)

**Read delta file from Lakehouse, to Spark DataFrame,**

**Then we need to convert it to Pandas DataFrame, and then we can use various Pandas functions to manipulate values for our test**

```
In [4]: source_path = "Files/staging/customers"

df_source = spark.read.format("delta").load(source_path)
```

```
StatementMeta(, 9d98d0b2-e722-4853-81cd-23f0968cf40f, 6, Finished, Available, Finished, False)
```

```
In [5]: df_source.orderBy("customer_id").show(40)
```

```
StatementMeta(, 9d98d0b2-e722-4853-81cd-23f0968cf40f, 7, Finished, Available, Finished, False)
```

| +-----+-----+-----+-----+-----+ |                     |                 |            |                      |
|---------------------------------|---------------------|-----------------|------------|----------------------|
| -----+                          |                     |                 |            |                      |
| customer_id                     | customer_name       | customer_status | address    | effective_start      |
| is_current                      |                     |                 |            |                      |
| +-----+-----+-----+-----+-----+ |                     |                 |            |                      |
| -----+                          |                     |                 |            |                      |
| C0001                           | Customer_0001       | Active          | 1 Main St  | 2026-02-04 23:14:... |
| true                            |                     |                 |            |                      |
| C0002                           | Customer_0002       | Inactive        | 2 Main St  | 2026-02-04 23:14:... |
| true                            |                     |                 |            |                      |
| C0003                           | Customer_0003       | Active          | 3 Main St  | 2026-02-04 23:14:... |
| true                            |                     |                 |            |                      |
| C0004                           | Customer_0004       | Active          | 4 Main St  | 2026-02-04 23:14:... |
| true                            |                     |                 |            |                      |
| C0005                           | Customer_0005_v2_v2 | Inactive        | 520 New St | 2026-02-04 23:14:... |
| true                            |                     |                 |            |                      |
| C0006                           | Customer_0006_v2_v2 | Inactive        | 976 New St | 2026-02-04 23:14:... |
| true                            |                     |                 |            |                      |
| C0007                           | Customer_0007       | Active          | 7 Main St  | 2026-02-04 23:14:... |
| true                            |                     |                 |            |                      |
| C0008                           | Customer_0008       | Inactive        | 8 Main St  | 2026-02-04 23:14:... |
| true                            |                     |                 |            |                      |
| C0009                           | Customer_0009       | Active          | 9 Main St  | 2026-02-04 23:14:... |
| true                            |                     |                 |            |                      |
| C0010                           | Customer_0010       | Active          | 10 Main St | 2026-02-04 23:14:... |
| true                            |                     |                 |            |                      |
| C0011                           | Customer_0011       | Active          | 11 Main St | 2026-02-04 23:14:... |
| true                            |                     |                 |            |                      |
| C0012                           | Customer_0012       | Inactive        | 12 Main St | 2026-02-04 23:14:... |
| true                            |                     |                 |            |                      |
| C0013                           | Customer_0013       | Inactive        | 13 Main St | 2026-02-04 23:14:... |
| true                            |                     |                 |            |                      |
| C0014                           | Customer_0014       | Active          | 14 Main St | 2026-02-04 23:14:... |
| true                            |                     |                 |            |                      |
| C0015                           | Customer_0015       | Active          | 15 Main St | 2026-02-04 23:14:... |
| true                            |                     |                 |            |                      |
| C0016                           | Customer_0016       | Active          | 16 Main St | 2026-02-04 23:14:... |
| true                            |                     |                 |            |                      |
| C0017                           | Customer_0017       | Active          | 17 Main St | 2026-02-04 23:14:... |
| true                            |                     |                 |            |                      |
| C0018                           | Customer_0018       | Active          | 18 Main St | 2026-02-04 23:14:... |
| true                            |                     |                 |            |                      |
| C0019                           | Customer_0019       | Active          | 19 Main St | 2026-02-04 23:14:... |
| true                            |                     |                 |            |                      |
| C0020                           | Customer_0020       | Active          | 20 Main St | 2026-02-04 23:14:... |
| true                            |                     |                 |            |                      |
| C0021                           | Customer_0021_v2_v2 | Active          | 394 New St | 2026-02-04 23:14:... |
| true                            |                     |                 |            |                      |
| C0022                           | Customer_0022       | Active          | 22 Main St | 2026-02-04 23:14:... |
| true                            |                     |                 |            |                      |
| C0023                           | Customer_0023       | Active          | 23 Main St | 2026-02-04 23:14:... |
| true                            |                     |                 |            |                      |
| C0024                           | Customer_0024       | Active          | 24 Main St | 2026-02-04 23:14:... |
| true                            |                     |                 |            |                      |
| C0025                           | Customer_0025       | Active          | 25 Main St | 2026-02-04 23:14:... |
| true                            |                     |                 |            |                      |
| C0026                           | Customer_0026_v2_v2 | Inactive        | 263 New St | 2026-02-04 23:14:... |
| true                            |                     |                 |            |                      |
| C0027                           | Customer_0027       | Active          | 27 Main St | 2026-02-04 23:14:... |
| true                            |                     |                 |            |                      |

|      |       |                     |          |            |                      |
|------|-------|---------------------|----------|------------|----------------------|
| true | C0028 | Customer_0028       | Active   | 28 Main St | 2026-02-04 23:14:... |
| true | C0029 | Customer_0029       | Active   | 29 Main St | 2026-02-04 23:14:... |
| true | C0030 | Customer_0030       | Active   | 30 Main St | 2026-02-04 23:14:... |
| true | C0031 | Customer_0031       | Active   | 31 Main St | 2026-02-04 23:14:... |
| true | C0032 | Customer_0032       | Active   | 32 Main St | 2026-02-04 23:14:... |
| true | C0033 | Customer_0033       | Active   | 33 Main St | 2026-02-04 23:14:... |
| true | C0034 | Customer_0034_v2_v2 | Inactive | 643 New St | 2026-02-04 23:14:... |
| true | C0035 | Customer_0035       | Inactive | 35 Main St | 2026-02-04 23:14:... |
| true | C0036 | Customer_0036       | Inactive | 36 Main St | 2026-02-04 23:14:... |
| true | C0037 | Customer_0037       | Active   | 37 Main St | 2026-02-04 23:14:... |
| true | C0038 | Customer_0038       | Active   | 38 Main St | 2026-02-04 23:14:... |
| true | C0039 | Customer_0039       | Active   | 39 Main St | 2026-02-04 23:14:... |
| true | C0040 | Customer_0040       | Active   | 40 Main St | 2026-02-04 23:14:... |

+-----+-----+-----+-----+-----+  
 -----+  
 only showing top 40 rows

## Convert Spark DataFrame to Pandas DataFrame

```
In [6]: pdf = df_source.toPandas()
```

```
StatementMeta(, 9d98d0b2-e722-4853-81cd-23f0968cf40f, 8, Finished, Available, Finished, False)
```

Make random changes now to the columns we will be tracking for our SCD Type 2 in Pandas DataFrame, like:

```
In [7]: for idx in rows_to_change:
        # Change customer_status randomly
        pdf.loc[idx, "customer_status"] = np.random.choice(["Active", "Inactive"])

        # Change address slightly
        pdf.loc[idx, "address"] = f"{random.randint(100,999)} New St"

        # Optionally change name for testing
        pdf.loc[idx, "customer_name"] = f"{pdf.loc[idx, 'customer_name']}_v2"
```

```
StatementMeta(, 9d98d0b2-e722-4853-81cd-23f0968cf40f, 9, Finished, Available, Finished, False)
```

Reset the dates and tracking flags once we've done our changes for the SCD scenario we are testing for in the dimension

```
In [8]: from datetime import datetime

        pdf["effective_start"] = datetime.now()
```

```
pdf["effective_end"] = None  
pdf["is_current"] = True
```

StatementMeta(, 9d98d0b2-e722-4853-81cd-23f0968cf40f, 10, Finished, Available, Finished, False)

## Convert back to Spark DataFrame

```
In [9]: df_source_modified = spark.createDataFrame(pdf)
```

StatementMeta(, 9d98d0b2-e722-4853-81cd-23f0968cf40f, 11, Finished, Available, Finished, False)

## Save to your staging folder

```
In [10]: source_path = "Files/staging/customers"  
df_source_modified.write.format("delta").mode("overwrite").save(source_path)
```

StatementMeta(, 9d98d0b2-e722-4853-81cd-23f0968cf40f, 12, Finished, Available, Finished, False)

**Let's view what changes we've done for our SCD test**

```
In [11]: df_source_modified.orderBy("customer_id").show(40)
```

StatementMeta(, 9d98d0b2-e722-4853-81cd-23f0968cf40f, 13, Finished, Available, Finished, False)

| +-----+-----+-----+-----+-----+ |                      |                 |            |                      |
|---------------------------------|----------------------|-----------------|------------|----------------------|
| +-----+-----+                   |                      |                 |            |                      |
| customer_id                     | customer_name        | customer_status | address    | effective_start      |
| is_current                      | effective_end        |                 |            |                      |
| +-----+-----+-----+-----+-----+ |                      |                 |            |                      |
| +-----+-----+                   |                      |                 |            |                      |
| C0001                           | Customer_0001        | Active          | 1 Main St  | 2026-02-05 00:11:... |
| true                            | NULL                 |                 |            |                      |
| C0002                           | Customer_0002        | Inactive        | 2 Main St  | 2026-02-05 00:11:... |
| true                            | NULL                 |                 |            |                      |
| C0003                           | Customer_0003        | Active          | 3 Main St  | 2026-02-05 00:11:... |
| true                            | NULL                 |                 |            |                      |
| C0004                           | Customer_0004        | Active          | 4 Main St  | 2026-02-05 00:11:... |
| true                            | NULL                 |                 |            |                      |
| C0005                           | Customer_0005_v2_v2  | Inactive        | 520 New St | 2026-02-05 00:11:... |
| true                            | NULL                 |                 |            |                      |
| C0006                           | Customer_0006_v2_v2  | Inactive        | 976 New St | 2026-02-05 00:11:... |
| true                            | NULL                 |                 |            |                      |
| C0007                           | Customer_0007        | Active          | 7 Main St  | 2026-02-05 00:11:... |
| true                            | NULL                 |                 |            |                      |
| C0008                           | Customer_0008        | Inactive        | 8 Main St  | 2026-02-05 00:11:... |
| true                            | NULL                 |                 |            |                      |
| C0009                           | Customer_0009_v2     | Active          | 501 New St | 2026-02-05 00:11:... |
| true                            | NULL                 |                 |            |                      |
| C0010                           | Customer_0010        | Active          | 10 Main St | 2026-02-05 00:11:... |
| true                            | NULL                 |                 |            |                      |
| C0011                           | Customer_0011        | Active          | 11 Main St | 2026-02-05 00:11:... |
| true                            | NULL                 |                 |            |                      |
| C0012                           | Customer_0012        | Inactive        | 12 Main St | 2026-02-05 00:11:... |
| true                            | NULL                 |                 |            |                      |
| C0013                           | Customer_0013        | Inactive        | 13 Main St | 2026-02-05 00:11:... |
| true                            | NULL                 |                 |            |                      |
| C0014                           | Customer_0014        | Active          | 14 Main St | 2026-02-05 00:11:... |
| true                            | NULL                 |                 |            |                      |
| C0015                           | Customer_0015        | Active          | 15 Main St | 2026-02-05 00:11:... |
| true                            | NULL                 |                 |            |                      |
| C0016                           | Customer_0016        | Active          | 16 Main St | 2026-02-05 00:11:... |
| true                            | NULL                 |                 |            |                      |
| C0017                           | Customer_0017        | Active          | 17 Main St | 2026-02-05 00:11:... |
| true                            | NULL                 |                 |            |                      |
| C0018                           | Customer_0018        | Active          | 18 Main St | 2026-02-05 00:11:... |
| true                            | NULL                 |                 |            |                      |
| C0019                           | Customer_0019        | Active          | 19 Main St | 2026-02-05 00:11:... |
| true                            | NULL                 |                 |            |                      |
| C0020                           | Customer_0020        | Active          | 20 Main St | 2026-02-05 00:11:... |
| true                            | NULL                 |                 |            |                      |
| C0021                           | Customer_0021_v2_... | Active          | 144 New St | 2026-02-05 00:11:... |
| true                            | NULL                 |                 |            |                      |
| C0022                           | Customer_0022        | Active          | 22 Main St | 2026-02-05 00:11:... |
| true                            | NULL                 |                 |            |                      |
| C0023                           | Customer_0023        | Active          | 23 Main St | 2026-02-05 00:11:... |
| true                            | NULL                 |                 |            |                      |
| C0024                           | Customer_0024        | Active          | 24 Main St | 2026-02-05 00:11:... |
| true                            | NULL                 |                 |            |                      |
| C0025                           | Customer_0025        | Active          | 25 Main St | 2026-02-05 00:11:... |
| true                            | NULL                 |                 |            |                      |
| C0026                           | Customer_0026_v2_v2  | Inactive        | 263 New St | 2026-02-05 00:11:... |
| true                            | NULL                 |                 |            |                      |
| C0027                           | Customer_0027        | Active          | 27 Main St | 2026-02-05 00:11:... |
| true                            | NULL                 |                 |            |                      |

